

Math Thermo

class 01

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Non-Equilibrium Thermodynamics

Course Plan

- 1) Equilibrium Thermodynamics
- 2) Statistical Mechanics
- 3) Kinetic Theory of Gases
- 4) Conservation laws in Continuous Systems
- 5) Entropy Production
- 6) Onsager's Principle
- 7) Applications

Equilibrium Thermodynamics

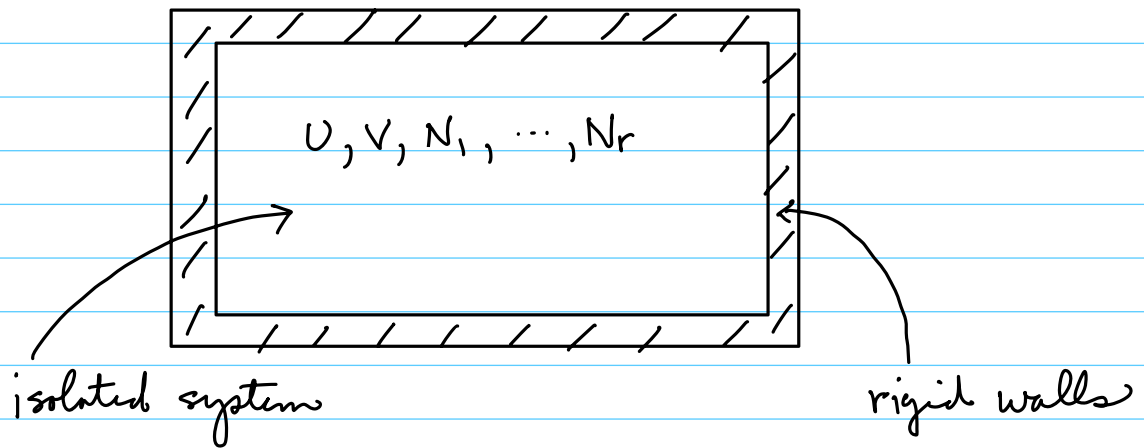
Defn (1.1): An isolated system is a collection of matter that shares no information with the outside world.

Postulate I: There are particular states, called equilibrium states, of an isolated system that, macroscopically, are characterized by

U , the internal energy of the system, $U \geq 0$,

V , the volume of the system, $V \geq 0$,

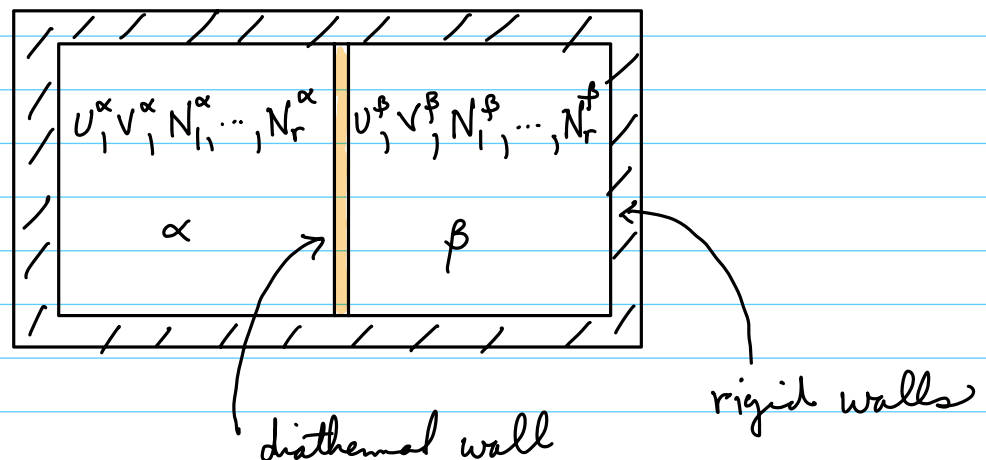
$N_1, \dots, N_r$, the mole numbers of the r chemical components of the system, $N_i \geq 0$.



The rigid walls that isolate the system from the rest of the universe allow no change in volume, no exchange of matter, and no exchange of energy.

Defn (1.2): A composite system is the union of two isolated systems that can exchange volume, matter (chemical components), and/or energy. However, no volume, matter, or energy is exchanged with the outside world.

A diathermal wall in a composite system is one which separates two otherwise isolated systems and allows for the exchange of energy, but not matter or volume.



A diathermal piston in a composite system is one which separates two otherwise isolated systems and allows for the exchange of energy and volume but not matter.

Postulate II: There exists a function $\tilde{S}$ for an isolated system called its entropy, defined for the system at equilibrium and depending on $U, V, N_1, \dots, N_r$, ie,

$$\tilde{S} = \tilde{S}(U, V, N_1, \dots, N_r).$$

The entropy exists for a composite system and is the sum of the entropy functions, $\tilde{S}^\alpha$ and $\tilde{S}^\beta$, for the respective subsystems, ie,

$$\tilde{S} = \tilde{S}^\alpha + \tilde{S}^\beta = \tilde{S}^\alpha(U^\alpha, V^\alpha, N_1^\alpha, \dots, N_r^\alpha) + \tilde{S}^\beta(U^\beta, V^\beta, N_1^\beta, \dots, N_r^\beta)$$

(1.1)

The entropy of an isolated system is homogeneous of degree 1, meaning,

$$\tilde{S}(\lambda U, \lambda V, \lambda N_1, \dots, \lambda N_r) = \lambda \tilde{S}(U, V, N_1, \dots, N_r)$$

for any $\lambda > 0$. (*extensive parameter*)

Postulate III: The entropy of an isolated system is a concave, twice continuously differentiable, *non-negative* function over its (convex) domain

$$\Sigma_U \subseteq [0, \infty)^{r+2} = [0, \infty) \times [0, \infty) \times [0, \infty) \times \dots \times [0, \infty)$$

(U) (V) (N₁) ... (N_r)

The entropy is a monotonically increasing function of U . In particular,

$$\left(\frac{\partial \tilde{S}}{\partial U}\right)_{V, N_1, \dots, N_r} > 0, \quad \forall (U, V, \vec{N}) \in \overset{\text{interior of } \Sigma_U}{\Sigma_U^\circ}.$$

Theorem (1.3): Suppose that Postulates I - III hold for an isolated system. Then, there is a function

$$\tilde{U} = \tilde{U}(S, V, N_1, \dots, N_r)$$

and a convex domain of definition

$$\Sigma_S = \underbrace{[0, \infty)}_{(S)}^{\uparrow+2} = \underbrace{[0, \infty)}_{(V)} \times \underbrace{[0, \infty)}_{(N_1)} \times \dots \times \underbrace{[0, \infty)}_{(N_r)}$$

that satisfies

$$\tilde{U}(\tilde{S}(U, V, \vec{N}), V, \vec{N}) = U, \quad \forall (U, V, \vec{N}) \in \Sigma_U$$

and

$$\tilde{S}(\tilde{U}(S, V, \vec{N}), V, \vec{N}) = S, \quad \forall (S, V, \vec{N}) \in \Sigma_S$$

Moreover, $\tilde{U}$ is a twice continuously differentiable, convex function with the property

$$\left(\frac{\partial \tilde{U}}{\partial S}\right)_{V, N_1, \dots, N_r} > 0, \quad \forall (S, V, \vec{N}) \in \Sigma_S.$$

Example (1.4): Suppose that, for a unary material ($r=1$),

$$\tilde{S}(U, V, N) = \left(\frac{NVUR^2}{v_0 \Theta} \right)^{1/3}, \quad \Sigma_U \in [0, \infty)^3,$$

where $R, v_0, \Theta > 0$ are constants. Consider the function

$$\tilde{U}(S, V, N) = \frac{S^3 v_0 \Theta}{NVR^2}, \quad \Sigma_S \in [0, \infty)^3.$$

Then

and $\tilde{S}(\tilde{U}(S, V, N), V, N) = S, \quad \forall (S, V, N) \in \Sigma_S$

$$\tilde{U}(\tilde{S}(U, V, N), V, N) = U, \quad \forall (U, V, N) \in \Sigma_U.$$

Defn (1.5): Suppose that Postulates I - III hold. Then the temperature of an equilibrium isolated system is defined as

$$(1.2) \quad T_U(S, V, N_1, \dots, N_r) \equiv \left(\frac{\partial \tilde{U}}{\partial S} \right)_{V, N_1, \dots, N_r}.$$

Postulate IV: Equilibrium of a composite system is that state $((U^\alpha, V^\alpha, \vec{N}^\alpha), (U^\beta, V^\beta, \vec{N}^\beta)) \in \Sigma_U^\alpha \times \Sigma_U^\beta$ such that

$$S = S^\alpha + S^\beta$$

is at its maximum possible value.

Defn (1.6): The function

$$(1.3) \quad P_v(s, v, N_1, \dots, N_r) := - \left(\frac{\partial \tilde{U}}{\partial v} \right)_{s, N_1, \dots, N_r}$$

is called the pressure of a system. The function

$$(1.4) \quad \mu_{v,i}(s, v, N_1, \dots, N_r) := \left(\frac{\partial \tilde{U}}{\partial N_i} \right)_{s, N_j \neq N_i}$$

is called the i^{th} chemical potential.

Theorem (1.7): let $\tilde{S}$ and $\tilde{U}$ be the entropy and internal energy functions of an isolated system. Then,

$$\frac{\partial \tilde{S}}{\partial U} = \frac{1}{T_s(u, v, \vec{N})},$$

where

$$T_u(s, v, \vec{N}) = T_s(\tilde{U}(s, v, N), v, \vec{N})$$

and

$$T_s(u, v, \vec{N}) = T_u(\tilde{S}(u, v, \vec{N}), v, \vec{N}).$$

Furthermore,

$$\frac{\partial \tilde{S}}{\partial v} = \frac{P_s(u, v, \vec{N})}{T_s(u, v, \vec{N})},$$

where

$$P_u(s, v, \vec{N}) = P_s(\tilde{U}(s, v, N), v, \vec{N}),$$

and

$$P_s(u, v, \vec{N}) = P_u(\tilde{S}(u, v, \vec{N}), v, \vec{N}).$$

And, finally,

$$\frac{\partial \tilde{S}}{\partial N_j} = - \frac{\mu_{s,j}(U, V, \vec{N})}{T_s(U, V, \vec{N})},$$

where

and $\mu_{v,j}(S, V, \vec{N}) = \mu_{s,j}(\tilde{U}(S, V, \vec{N}), V, \vec{N})$

$$\mu_{s,j}(U, V, N) = \mu_{v,j}(\tilde{S}(U, V, \vec{N}), V, \vec{N}).$$

Proof Exercise. //

Remark: We will usually abuse notation and just write

$$T = T_v = T_s, \quad P = P_v = P_s, \quad \mu_j = \mu_{v,j} = \mu_{s,j},$$

when the usage may inferred from the context.

Postulate V: The entropy of an isolated system is zero when the temperature is zero, that is,

$$T_s(U, V, N_1, \dots, N_r) = 0 \Rightarrow \tilde{S}(U, V, N_1, \dots, N_r) = 0.$$

Example (1.8): Consider a unary ($r=1$) isolated system in equilibrium with the fundamental relation

$$\tilde{S}(U, V, N) = \left(\frac{NVUR^2}{v_0 \Theta} \right)^{1/3}.$$

We can set

$$\Sigma_v = [0, \infty)^3.$$

$\tilde{S} = 0$ for all $(U, V, N) \in \partial \Sigma_v$, where

$$\partial \Sigma_v = \{(u, v, N) \in \Sigma_v \mid u=0, \text{ or } v=0, \text{ or } N=0\}$$

Recall,

$$\begin{aligned} T_s(u, v, N) &= \frac{3v_0 \theta}{NVR^2} \left(\tilde{S}(u, v, N) \right)^2 \\ &= \frac{3v_0 \theta}{NVR} \left(\frac{NVUR^2}{v_0 \theta} \right)^{2/3}. \end{aligned}$$

T_s is not defined for all of Σ_v ! It is defined for all

$$(u, v, N) \in \text{dom}(T_s) \subset \Sigma_v$$

Define

$$AZ := \{(u, v, N) \in \text{dom}(T_s) \mid T_s(u, v, N) = 0\}.$$

Then

$$AZ \subset \partial \Sigma_v. \quad (\text{Postulate V})$$

I leave it as a homework exercise to determine $\text{dom}(T_s)$ and AZ .